Lecture Notes V

Course: Analytical Mechanics

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Topic: Euler Angles

Abstract. In this section we will study a classical example of a rigid body, called the symmetric top and discuss the equations of motion (using the Euler angles as generalised coordinates) using the Hamiltonian formalism.

Symmetric Top

A rigid body is said to be **symmetric** if two of its principal moments of inertia coincide.

Example (Symmetric top in a uniform gravitational field). [1] Let us consider an axially symmetric top with a fixed point, in a uniform gravitational field. Assuming that the ξ_3 (in the body attached frame) axis is the symmetry axis, so that the principal moments of inertia I_1, I_2 coincide., assuming of course the inertia tensor is a diagonal matrix. The standard Lagrangian thus reads

$$L = T - V = \frac{1}{2}\omega_j^2 I_j - V.$$

In order to find the angular velocities ω_j let us parametrise the rotation of the spinning top by making use of the so-called **Euler angles**, which are defined as $(\psi, \theta, \phi), 0 \leq \psi < 2\pi, 0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi$ by means of the rotation matrix A:

$$\mathsf{A} = \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\mathsf{R}_z(\psi)$$

describing the motion of an object. Here, R_d is a rotation matrix about the d-axis. Just like we saw in the discussion on rigid bodies, the angular velocities ω_j are defined through

$$(\Omega)_{ij} = -\varepsilon_{ijk}\omega_k, \quad \dot{\mathsf{A}}\mathsf{A}^{\mathsf{T}} = \Omega = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}, \tag{1}$$

so we compute the matrix $\dot{\mathsf{A}}\mathsf{A}^{\mathsf{T}}$.

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{A} \equiv \dot{\mathsf{A}} = \dot{\mathsf{R}}_z(\phi)\mathsf{R}_x(\theta)\mathsf{R}_z(\psi) + \mathsf{R}_z(\phi)\dot{\mathsf{R}}_x(\theta)\mathsf{R}_z(\psi) + \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\dot{\mathsf{R}}_z(\psi), \quad \mathsf{A}^{\mathsf{T}} = \mathsf{R}_z^{\mathsf{T}}(\psi)\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)$$

$$\begin{aligned}
\dot{\mathsf{A}}\mathsf{A}^{\mathsf{T}} &= \left\{\dot{\mathsf{R}}_z(\phi)\mathsf{R}_x(\theta)\mathsf{R}_z(\psi) + \mathsf{R}_z(\phi)\dot{\mathsf{R}}_x(\theta)\mathsf{R}_z(\psi) + \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\dot{\mathsf{R}}_z(\psi)\right\}\left[\mathsf{R}_z^{\mathsf{T}}(\psi)\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)\right] \\
&= \dot{\mathsf{R}}_z(\phi)\underbrace{\mathsf{R}_x(\theta)\mathsf{R}_z(\psi)\mathsf{R}_z^{\mathsf{T}}(\psi)\mathsf{R}_x^{\mathsf{T}}(\theta)}_{\mathsf{I}}\mathsf{R}_z^{\mathsf{T}}(\phi) + \mathsf{R}_z(\phi)\dot{\mathsf{R}}_x(\theta)\underbrace{\mathsf{R}_z(\psi)\mathsf{R}_z^{\mathsf{T}}(\psi)}_{\mathsf{I}}\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi) + \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\dot{\mathsf{R}}_z(\psi)\mathsf{R}_z^{\mathsf{T}}(\psi)\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi) \\
&= \Omega_z^{\phi} + \mathsf{R}_z(\phi)\Omega_x^{\theta}\mathsf{R}_z^{\mathsf{T}}(\phi) + \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\Omega_z^{\psi}\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)
\end{aligned}$$

where we defined $\Omega_d^{\alpha} \equiv \dot{\mathsf{R}}_d(\alpha)\mathsf{R}_d^{\mathsf{T}}(\alpha)$ and used the fact that the rotation matrices R_d are orthogonal, i.e., $\mathsf{R}_d\mathsf{R}_d^{\mathsf{T}} = \mathsf{I}$.

$$\dot{\mathsf{R}}_z(\phi) = \frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \dot{\phi}\begin{pmatrix} -\sin\phi & -\cos\phi & 0 \\ \cos\phi & -\sin\phi & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathsf{R}_z^{\mathsf{T}}(\phi) = \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Omega_z^{\phi} = \begin{pmatrix} 0 & -\dot{\phi} & 0 \\ \dot{\phi} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\dot{\mathsf{R}}_x(\theta)=\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}1&0&0\\0&\cos\theta&-\sin\theta\\0&\sin\theta&\cos\theta\end{pmatrix}=\dot\theta\begin{pmatrix}0&0&0\\0&-\sin\theta&-\cos\theta\\0&\cos\theta&-\sin\theta\end{pmatrix},\quad \mathsf{R}_x^{\mathsf{T}}(\theta)=\begin{pmatrix}1&0&0\\0&\cos\theta&\sin\theta\\0&-\sin\theta&\cos\theta\end{pmatrix}$$

$$\Omega_x^\theta=\begin{pmatrix}0&0&0\\0&0&-\dot\theta\\0&\dot\theta&0\end{pmatrix}$$

$$\Omega_z^\psi=\begin{pmatrix}0&-\dot\psi&0\\\dot\psi&0&0\\0&0&0\end{pmatrix}$$

$$\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)=\begin{pmatrix}1&0&0\\0&\cos\theta&\sin\theta\\0&-\sin\theta&\cos\theta\end{pmatrix}\begin{pmatrix}\cos\phi&\sin\phi&0\\-\sin\phi&\cos\phi&0\\0&0&1\end{pmatrix}=\begin{pmatrix}\cos\phi&\sin\phi&0\\-\sin\phi\cos\theta&\cos\phi\cos\theta&\sin\theta\\\sin\phi\sin\theta&-\cos\phi\sin\theta&\cos\theta\end{pmatrix}$$

$$\Omega_z^\psi\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)=\dot\psi\begin{pmatrix}\sin\phi\cos\theta&-\cos\phi\cos\theta&-\sin\theta\\\cos\phi&\sin\phi&0\\0&0&0\end{pmatrix},\quad \mathsf{R}_z(\phi)\mathsf{R}_x(\theta)=\begin{pmatrix}\cos\phi&-\sin\phi\cos\theta&\sin\phi\sin\theta\\\sin\phi&\cos\phi\cos\theta&-\cos\phi\sin\theta\\0&\sin\theta&\cos\theta\end{pmatrix}$$

$$\mathsf{R}_z(\phi)\mathsf{R}_x(\theta)\Omega_z^\psi\mathsf{R}_x^{\mathsf{T}}(\theta)\mathsf{R}_z^{\mathsf{T}}(\phi)=\dot\psi\begin{pmatrix}0&-\cos\theta&\sin\theta\cos\phi\\\cos\theta&0&-\sin\theta\sin\phi\\\cos\phi\sin\theta&\sin\phi\sin\theta&0\end{pmatrix}\qquad(*)$$

$$\mathsf{R}_z(\phi)\Omega_x^\theta=\dot\theta\begin{pmatrix}0&0&\sin\phi\\0&0&-\cos\phi\\0&1&0\end{pmatrix}$$

$$\mathsf{R}_z(\phi)\Omega_x^\theta\mathsf{R}_z^{\mathsf{T}}(\phi)=\dot\theta\begin{pmatrix}0&0&\sin\phi\\0&0&-\cos\phi\\-\sin\phi&\cos\phi&0\end{pmatrix}\qquad(**)$$

adding $(*)$, $(**)$ and Ω_z^ϕ we get

$$\dot{\mathsf{A}}\mathsf{A}^{\mathsf{T}}=\begin{pmatrix}0&-\dot\psi\cos\theta-\dot\phi&-\dot\psi\sin\theta\cos\phi+\dot\theta\sin\phi\\\dot\psi\cos\theta+\dot\phi&0&-\dot\psi\sin\theta\sin\phi-\dot\theta\cos\phi\\\dot\psi\cos\phi\sin\theta-\dot\theta\sin\phi&\dot\psi\sin\phi\sin\theta+\dot\theta\cos\phi&0\end{pmatrix}=\Omega\qquad(2)$$

so that on comparing (2) with (1) we get

$$\begin{aligned}\omega_1&=\dot\psi\sin\phi\sin\theta+\dot\theta\cos\phi,\\\omega_2&=\dot\theta\sin\phi-\dot\psi\sin\theta\cos\phi,\\\omega_3&=\dot\phi+\dot\psi\cos\theta.\end{aligned}$$

This way, using that $I_1=I_2$ the Lagrangian reads

$$L(\phi,\dot\phi,\theta,\dot\theta,\psi,\dot\psi)=\frac{1}{2}I_1\left(\dot\theta^2+\dot\psi^2\sin^2\theta\right)+\frac{1}{2}I_3(\dot\psi\cos\theta+\dot\phi),^2-Mgl\cos\theta$$

and then we can compute the canonical momenta

$$p_\phi=I_3(\dot\psi\cos\theta+\dot\phi),\quad p_\theta=I_1\dot\theta,\quad p_\psi=I_1\dot\psi\sin^2\theta+I_3\cos\theta(\dot\psi\cos\theta+\dot\phi)\qquad(3)$$

so that the Hamiltonian reads

$$H(\theta,p_\theta,\phi,p_\phi,\psi,p_\psi)=p_\theta\dot\theta+p_\phi\dot\phi+p_\psi\dot\psi-L$$

$$=I_1\dot\theta^2+I_3(\dot\psi\cos\theta+\dot\phi)\dot\phi+I_1\dot\psi^2\sin^2\theta+I_3\cos\theta(\dot\psi\cos\theta+\dot\phi)\dot\psi-\frac{1}{2}I_1\left(\dot\theta^2+\dot\psi^2\sin^2\theta\right)-\frac{1}{2}I_3(\dot\psi\cos\theta+\dot\phi)^2+\underbrace{Mgl\cos\theta}_{V}$$

$$= \frac{1}{2}I_1\dot{\theta}^2 + I_3(\dot{\psi}\cos\theta + \dot{\phi})\dot{\phi} + \frac{1}{2}I_1\dot{\psi}^2\sin^2\theta + I_3\cos\theta(\dot{\psi}\cos\theta + \dot{\phi})\dot{\psi} - \frac{1}{2}I_3\dot{\psi}^2\cos^2\theta - I_3\dot{\psi}\dot{\phi}\cos\theta - \frac{1}{2}I_3\dot{\phi}^2 + V$$

$$= \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_1\dot{\psi}^2\sin^2\theta + \underbrace{I_3\dot{\psi}\dot{\phi}\cos\theta}_{a} + I_3\dot{\phi}^2 + I_3\cos^2\theta\dot{\psi}^2 + I_3\cos\theta\dot{\psi}\dot{\phi} - \frac{1}{2}I_3\dot{\psi}^2\cos^2\theta - \underbrace{I_3\dot{\psi}\dot{\phi}\cos\theta}_{a} - \frac{1}{2}I_3\dot{\phi}^2 + V$$

$$= \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_1\dot{\psi}^2\sin^2\theta + \frac{1}{2}I_3\dot{\phi}^2 + I_3\cos\theta\dot{\psi}\dot{\phi} + \frac{1}{2}I_3\cos^2\theta\dot{\psi}^2 + V$$

$$= \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_3\dot{\phi}^2 + \frac{1}{2}I_1\dot{\psi}^2\sin^2\theta + \frac{1}{2}I_3\left[\left(\dot{\psi}\cos\theta + \dot{\phi}\right)^2 - \dot{\phi}^2\right] = \frac{1}{2}I_1\dot{\theta}^2 + \frac{I_1}{2}\dot{\psi}^2\sin^2\theta + \frac{1}{2}I_3\left(\dot{\psi}\cos\theta + \dot{\phi}\right)^2 + V$$

which on using (3) we see that $\dot{\phi} + \dot{\psi}\cos\theta = p_\phi/I_3$ and $\dot{\psi} = (p_\psi - p_\phi\cos\theta)/I_1\sin^2\theta$ so

$$H = \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_1\frac{(p_\psi - p_\phi\cos\theta)^2}{I_1^2\sin^4\theta}\sin^2\theta + \frac{1}{2}I_3\frac{p_\phi^2}{I_3^2} + Mgl\cos\theta,$$

now, we recall that it is convenient to express H in terms of p_θ, p_ϕ, p_ψ, and hence we have

$$H(\theta, p_\theta, \phi, p_\phi, \psi, p_\psi) = \frac{p_\theta^2}{2I_1} + \frac{(p_\psi - p_\phi\cos\theta)^2}{2I_1\sin^2\theta} + \frac{p_\phi^2}{2I_3} + Mgl\cos\theta$$

but looking at the Lagrangian one can see that p_ψ, p_ϕ are constants of the motion which we denote by a, b, respectively so that the Hamiltonian can be written as

$$H(\theta, p_\theta) = \frac{p_\theta^2}{2I_1} + \frac{(a - b\cos\theta)^2}{2I_1\sin^2\theta} + \frac{b^2}{I_3} + Mgl\cos\theta = \frac{p_\theta^2}{2I_1} + V_{eff}(\theta)$$

with $V_{eff}(\theta) = (a - b\cos\theta)^2/(2I_1\sin^2\theta) + b^2/I_3 + Mgl\cos\theta$, so we can now use the fact that since the Hamiltonian does not depend explicitly on time, then it is a constant of the motion (recall that for the Hamiltonian, one has $\mathrm{d}H/\mathrm{d}t = \partial H/\partial t$) denoted by E (this coincides with the energy because the term T in the Lagrangian is a homogeneous function of order 2 in the generalised velocities $(\dot{\psi}, \dot{\theta}, \dot{\phi})$) to reduce the problem to a fist-order ODE

$$E = \frac{1}{2}I_1\dot{\theta}^2 + \frac{(a - b\cos\theta)^2}{2I_1\sin^2\theta} + \frac{b^2}{I_3} + Mgl\cos\theta$$

References

[1] G. F. Torres del Castillo, *An Introduction to Hamiltonian Mechanics*, Birkhäuser, (2018).